



NO EMPTY PLATE

ETHOS

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Problem Statement

Rising Food prices in the United States make it harder for low-income households to afford nutritious groceries.

In 2025, families have paid approximately \$310 more on groceries compared with 2024 for essential items, such as milk, chicken and bananas (Fink, 2026).

These families spend a larger share of their income on food and often have to choose cheaper, less healthy options or skip essential items. Many communities also face uneven access to affordable grocery stores.

[scan QR for full citation]

Background Information

[USDA Food Access Research Atlas]
[Online Grocery Marketplaces]
[Grocery Price Search & Comparison Apps]

[scan QR to learn more]

Available Data

Grocery store locations
Census income & access data
Some price tracking data via apps
SNAP/EBT participation lists

Limited Data

Long-term price volatility at store level, nutrition quality across full inventories, real-time store comparisons of nutrition and price and equity.

Goals and Objectives

Pre-Design Steps (OC Based)

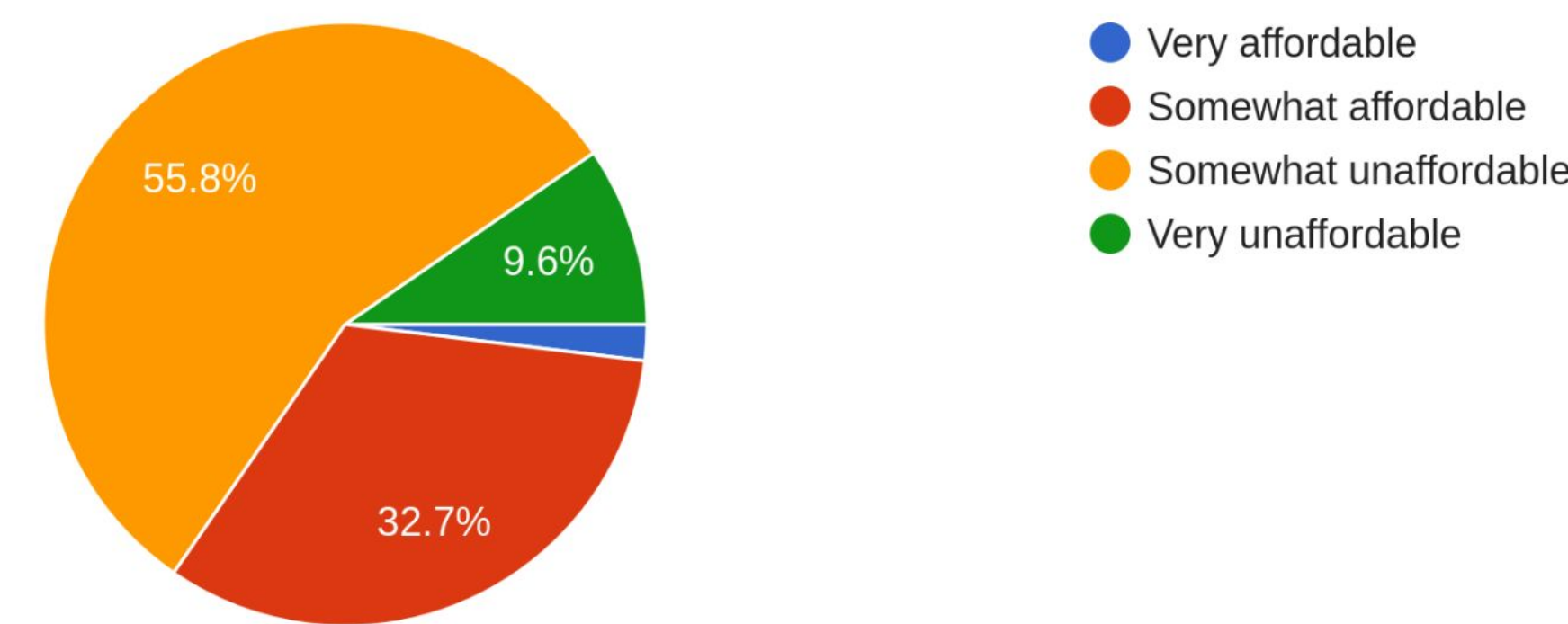
Collect grocery prices, nutrition info, and store locations.
Map neighborhoods for SNAP/EBT participation.

Post-Implementation Vision

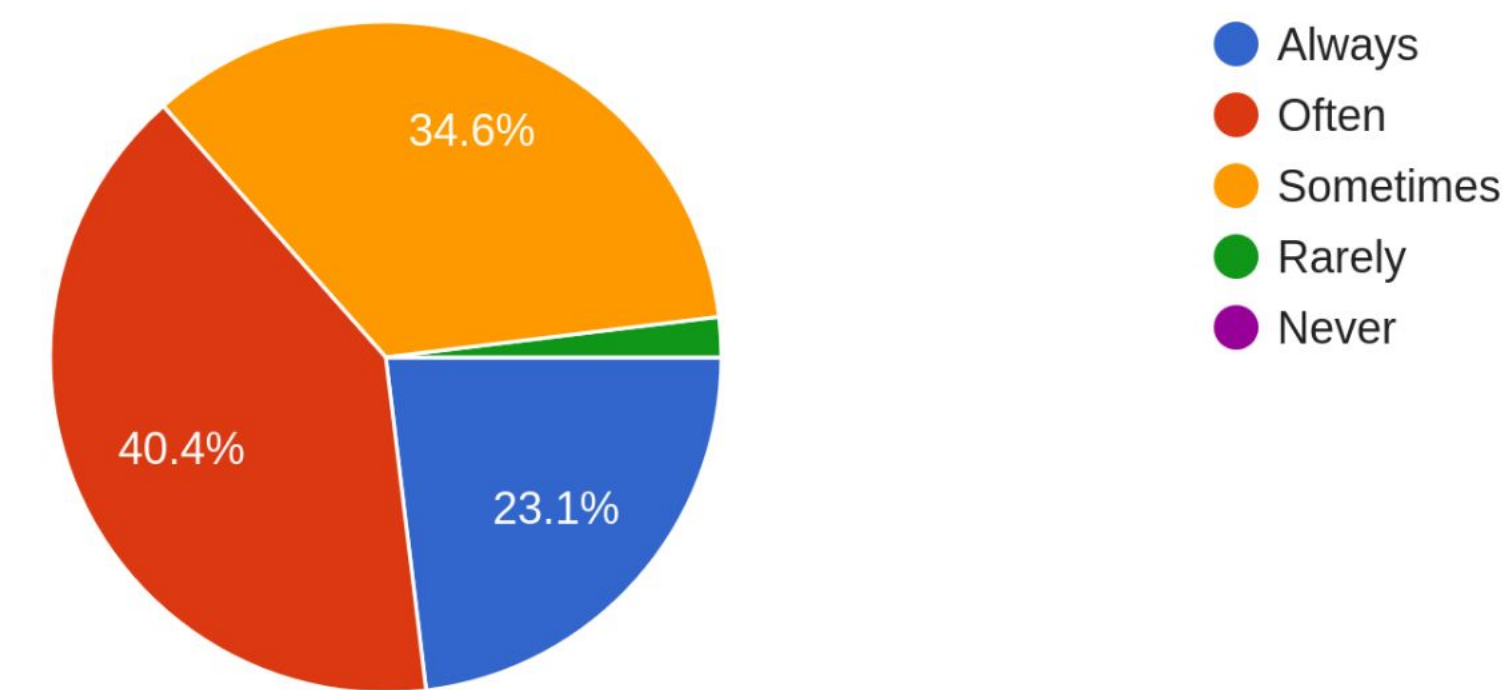
Low-income households in OC shall quickly find affordable and nutritious groceries from stores nearby.
Grocery systems shall become more transparent and reliable for all OC neighborhoods.

Grocery Equity Survey

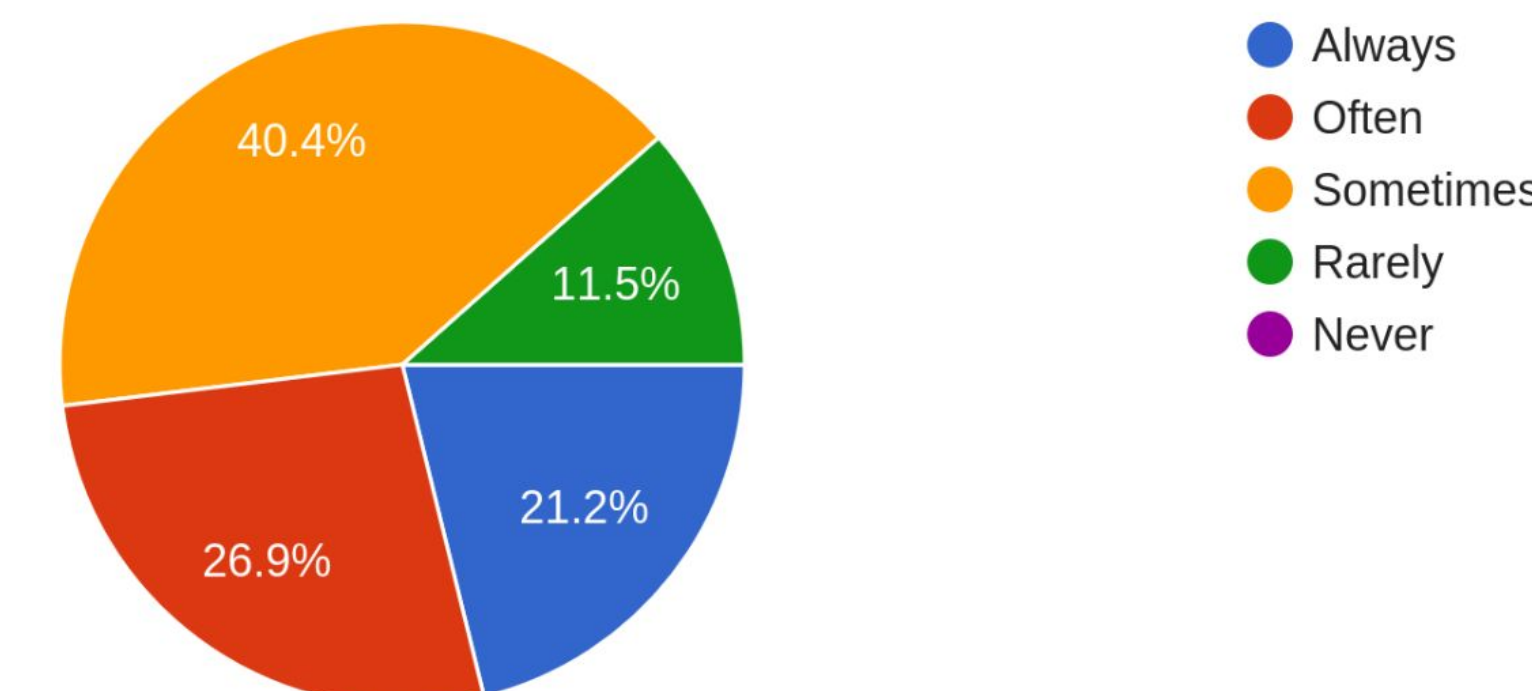
How affordable do you feel groceries are for you right now?
52 responses



How often do grocery price changes affect what you buy?
52 responses



Do you stock up on food when prices are lower?
52 responses



[Scan QR to view full survey]

What People Have to Say

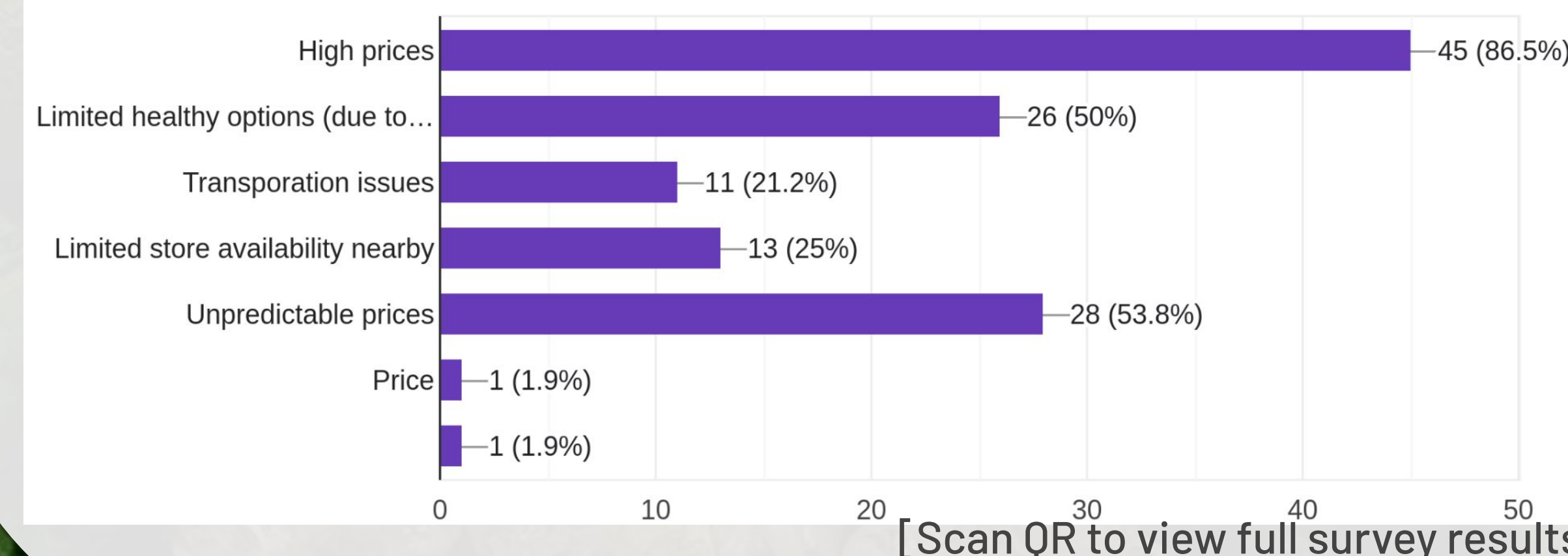
"Sometime I skip buying certain items because of my budget."

"I try to eat healthy but I usually end up choosing what's cheaper especially for snacks and protein."

"I usually stick to basic items and avoid specialty foods because they cost too much"

"I don't know which grocery stores offer the best prices"

Which of the following grocery-related challenges do you experience? (Select all that apply)
52 responses



[Scan QR to view full survey results]

Design Idea

- Each grocery store in OC is assigned an Equity Score based on their affordability, nutrition access, price volatility and participation in government food programs.
- An Equity Score ranges from 0 to 10 with 0 being least affordable and 10 being most affordable.
- Users can also access side-by-side store comparisons, price trends over time and distance/travel time to stores.
- The OC neighborhood Equity Score is calculated as the average of all store scores within that area.

Evidence of Future Success

86.5% of survey respondents lack affordable grocery access.
NEP connects food resources directly to them.

51.9% find nutrition and price equally important
NEP helps find a balance between nutrition and budget.

67.3% travel between 1 - 5 miles for groceries
NEP highlights neighborhoods with most affordable stores for when moving apartments.

Future Direction

[Nationwide expansion]

[Working alongside government]

↓ [Additional features] ↓

Create grocery list

Discount/sale alert

Food donation/events alert

View specific coupon eligible items

ETHOS

- **Sheikh Sabah Ali:** an enthusiastic Computer Science senior, who is inspired by his own family's struggles with affordable groceries.
- **Yu-Chen Chung:** a passionate Computer Science senior inspired to use technology to solve real-world problems.

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